Synergistic Fusion of Multitemporal and Multiresolution SAR Data: A Hierarchical Prior Transfer Approach for Aircraft Detection

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***Abstract*—Current deep learning methods for synthetic aperture radar (SAR) aircraft detection typically rely on single-temporal, high-resolution imagery, requiring extensive labeled datasets and limiting generalization across sensors and resolutions. Meanwhile, abundant multitemporal, coarse- and moderate-resolution SAR images from satellites such as Sentinel-1, offering valuable temporal cues about stable aircraft positions at designated parking stands, remain largely underutilized. In addition, existing techniques inadequately exploit the complementary advantages of multiresolution SAR data, overlooking the synergy between global context from coarse imagery and detailed localization from fine-resolution imagery. To bridge this gap, we propose the hierarchical prior transfer approach (HiPTA), a novel framework for synergistic fusion of multitemporal and multiresolution SAR data in a coarse-to-fine detection paradigm. First, the coarse-resolution multitemporal prior extraction (CMPE) module mines low-resolution time series to identify robust parking-stand priors. Next, the intermediate-resolution prior fusion and refinement (IPFR) and hierarchical multiresolution prior registration (HMPR) modules align and refine these priors across intermediate and high-resolution domains, drastically narrowing the search space. Finally, the fine-resolution aircraft detection (FRAD) module employs a domain-adaptive frequency-adaptive filtering network to classify candidate regions in high-resolution imagery. Extensive experiments on multisource C- and Ku-band SAR datasets show that HiPTA consistently improves precision-recall tradeoffs and maintains robustness across sensors and resolutions, delivering consistent gains over single-resolution detectors and airport-detection-aided methods.**

***Index Terms*—Aircraft detection, multiresolution, multitemporal, parking-stand, prior knowledge, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) is an active microwave imaging technology capable of capturing high-resolution images regardless of weather conditions or time of day. Owing to its unique imaging mechanism and extensive coverage, SAR has been widely applied across various domains [1]. Among these, the detection of aircraft in SAR imagery has gained particular prominence due to its critical importance in airport operations, air traffic management, and emergency response planning [2], [3].

Aircraft detection in SAR imagery remains challenging due to the inherent complexity of SAR data, including discrete scattering characteristics, background clutter, and imaging variability. Traditional methods relied heavily on background clutter modeling or hand-crafted feature extraction [4], [5], [6], but these techniques often fail to effectively address the complexities of modern SAR imagery. Recent advances have focused on leveraging the power of deep learning to improve detection performance. One notable approach is the enhancement of scattering features. For instance, Guo et al. [7] introduced a scattering-enhanced attention pyramid network that enhances scattering features to improve detection accuracy in challenging SAR environments. Similarly, Kang et al. [8] proposed the scattering feature relation network, which strengthens the relationships between scattering points to produce more accurate detection results. Chen et al. [9] proposed a method using point features that highlight key semantic areas, eliminating the reliance on anchor boxes, and introduced an incentive attention feature fusion strategy to improve feature fusion efficiency. Meng et al. [10] proposed a scattering topology cue-based network that enhances aircraft discriminability by modeling aircraft with a scattering topology and incorporating a space fusion module and a channel attention module. These approaches underscore leveraging scattering characteristics to enhance aircraft detection accuracy. Cascaded-attention designs such as CVANet [11] progressively refine feature, channel, and pixel representations, providing a representative attention stack that enhances fine-detail modeling in remote-sensing imagery. Zhang et al. [12] proposed an adversarial cross-aware network that couples semi-supervised GAN-based augmentation with cross-aware feature learning to improve discriminability under limited supervision.

In addition, incorporating geospatial context has proven valuable for improving detection performance. Luo et al. [13]

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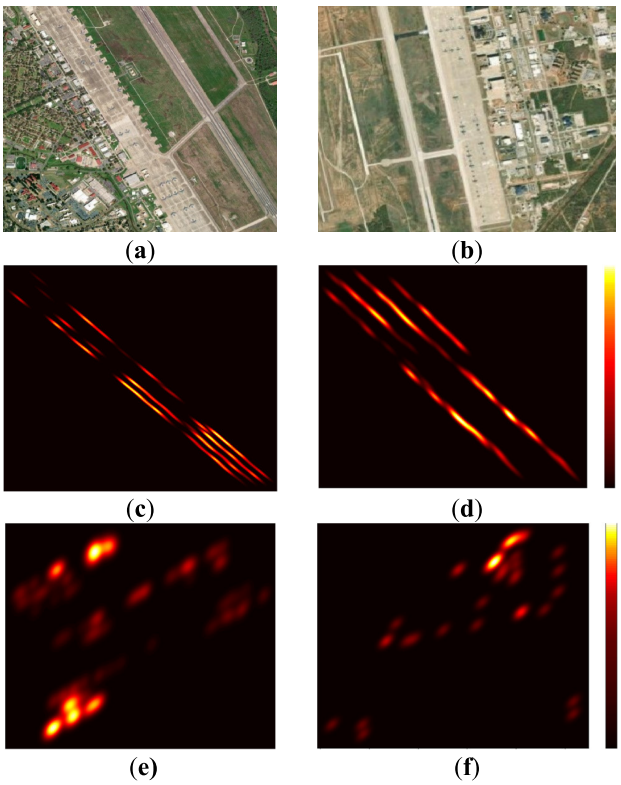

Fig. 1. Optical-SAR comparison of aircraft parking activity. (a) Optical image of Airport A. (b) optical image of Airport B. (c)–(f) Heatmaps of aircraft parking frequency derived from multitemporal SAR analysis for Airports A–D, respectively. Warmer colors indicate higher frequency; the color bars on the right show the normalized scale.

developed a geospatial contextual prior-enabled framework that uses airport locations to guide aircraft detection, providing valuable spatial cues to reduce false alarms and improve detection accuracy. Similarly, Chen et al. [14] introduced a geospatial transformer to process large-scale SAR imagery without exhaustive tiling, demonstrating that explicit spatial context improves aircraft detection in complex scenarios. The methods in [15] and [16] leverage prior geographic information for airport detection, improving aircraft detection efficiency and reducing false alarms.

Despite these advances, current SAR detection deep learning models typically rely on single-temporal, high-resolution SAR imagery for training and inference. They are often trained on a limited set of high-resolution images from a specific sensor, which makes them data-hungry and limits their generalizability [17], [18]. In practice, collecting large-scale labeled SAR datasets is challenging. This scarcity of annotations limits the applicability of deep neural networks, which typically require large amounts of training data to perform effectively [18]. As a result, models tend to overfit to the conditions seen in their training data. When these models are applied to images from different sensors or coarser resolutions, their performance can degrade significantly [19].

In this context, two underutilized opportunities emerge for improving detection performance and generalization. First, there is a wealth of lower resolution SAR imagery with high revisit frequency now available from satellite constellations. Sentinel-1's twin satellites, for instance, provide C-band SAR coverage of the Earth every six days, resulting in abundant multitemporal SAR image series [20]. While each Sentinel-1 image has moderate spatial resolution (~20 m), the high revisit rate enables time-series analysis of a scene [21]. By examining a sequence of SAR images over time, one can suppress random noise and isolate persistent scatterers or recurrent patterns [22]. In other words, multitemporal observations can be exploited to extract spatial priors about where targets of interest tend to appear. Modern airport operations adhere to well-defined logistical and regulatory frameworks, ensuring that aircraft remain in these carefully designed areas when on the ground [23], [24]. The visualization of aircraft parking frequency via a heatmap, as shown in Fig. 1, further illustrates this concept. The heatmaps for Airports A, B, C, and D depict the distribution of aircraft stop positions within these designated parking stands, highlighting the spatial patterns and frequencies of parked aircraft across the different airports. For example, recent studies demonstrate that analyzing a time series of Sentinel-1 images over airfields enables the detection of intermittent aircraft presence at parking stands [25]. Such temporal analysis effectively integrates information across acquisitions to highlight stable target signals or significant changes, providing cues that a single snapshot cannot offer. However, current deep learning detectors largely ignore multitemporal information—they operate on single temporal images, underutilizing the rich temporal context that could be gleaned from regularly updated SAR observations.

Second, modern approaches have not fully leveraged the synergy between multiresolution SAR data. Lower resolution SAR images and high-resolution SAR images offer complementary strengths. The coarse-resolution imagery covers large swaths and can capture global context: for instance, a cluster of aircraft on a tarmac might appear as a single bright blob in a low-res image, indicating general activity in that region [26]. In contrast, the fine resolution imagery provides local detail, resolving individual aircraft and exact shapes [27]. Intuitively, using a coarse image to find candidate regions and then analyzing those regions with a high-res image could yield a powerful coarse-to-fine detection strategy. This kind of cross-resolution collaboration is largely underexploited. Most existing methods process a single resolution at a time, often focusing exclusively on high-resolution data. When multiresolution data are considered, it is usually in a domain adaptation context rather than a direct information fusion [28]. In other words, current techniques treat low- and high-resolution SAR images as separate domains to bridge, instead of simultaneously exploiting the cross-resolution content for detection [29]. This leaves a significant gap—the cross-resolution synergy, where coarse images guide and complement fine-scale analysis, remains an untapped resource in today's SAR aircraft detection frameworks.

The above considerations motivate our proposed approach. To overcome the limitations of purely data-driven models, we seek to inject auxiliary information into the detection process. In particular, we leverage multitemporal, multiresolution SAR data to derive informative spatial priors and context that can aid in aircraft detection. Rather than training a deep model

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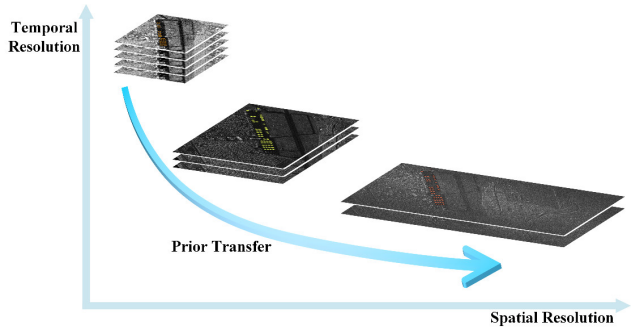

Fig. 2. Overview of the HiPTA. (Top-left) In particular, high temporal resolution yet coarse-spatial-resolution images are utilized first to extract robust prior knowledge of aircraft parking positions through multitemporal analysis. (Center) These priors are subsequently refined at intermediate-resolutions and (bottom-right) finally transferred to high spatial resolution but low-temporal-resolution SAR data.

in isolation on a limited high-res dataset, we augment it with knowledge gleaned from abundant lower resolution observations and temporal dynamics. By incorporating these priors, the detector can focus on likely target areas and account for typical background variations, thereby improving robustness and reducing the dependence on large labeled datasets. In this article, we propose a hierarchical prior transfer approach (HiPTA) that synergistically fuses multitemporal and multiresolution SAR information for aircraft detection. As illustrated in Fig. 2, our method leverages SAR images acquired from multiple satellites with complementary temporal and spatial coverage. First, high temporal frequency yet coarse-resolution satellite images provide abundant temporal cues to identify potential parking stands, forming initial priors. These are then refined at intermediate resolutions, where more precise spatial features and partial annotations help consolidate and adjust bounding boxes. Finally, the refined priors are transferred onto high spatial resolution but low-temporal-availability SAR data, ensuring that the detection process concentrates on the most relevant regions. Furthermore, at the fine-resolution level, refined priors guide the extraction of targeted image slices from high-resolution SAR data. These slices undergo keypoint-based feature extraction and classification through a frequency adaptive filtering network (FAFNet) integrated with domain adaptation, enabling effective aircraft detection.

The main contributions of our work can be summarized as follows.

1) A novel hierarchical prior transfer framework (HiPTA) is proposed for aircraft detection in SAR imagery. This framework synergistically fuses multitemporal and multiresolution SAR data in a coarse-to-fine manner, effectively leveraging broad temporal patterns alongside fine spatial details.

2) Multitemporal analysis is exploited to extract robust parking-stand priors from abundant low-resolution SAR data. By leveraging temporal variability in backscatter, candidate regions where aircraft are likely parked are automatically identified, providing essential guidance for subsequent detection stages.

3) A cross-resolution fusion strategy is developed to transfer and refine these coarse priors across intermediate and high-resolution SAR imagery. This approach aligns the spatial contexts derived from low-resolution data with the detailed features present in high-resolution images, thereby enhancing localization accuracy.

The remainder of this article is organized as follows. Section II provides the proposed HiPTA in detail. In Section III, the extensive experiments and analysis are presented. Finally, the conclusions are drawn in Section IV.

II. Methodology

A. Overall Architecture

In this section, we present the overall architecture of our proposed HiPTA, as illustrated in Fig. 3. It comprises four main modules: coarse-resolution multitemporal prior extraction (CMPE), intermediate-resolution prior fusion and refinement (IPFR), hierarchical multiresolution prior registration (HMPR), and fine-resolution aircraft detection (FRAD) with transferred priors. The approach leverages multitemporal and multiresolution SAR imagery to refine aircraft detection priors progressively in a coarse-to-fine manner. At the coarse-resolution level, multitemporal SAR images with high temporal frequency but low spatial resolution are analyzed to detect change points, prelocalize airport regions, and extract initial coarse parking-stand priors. Subsequently, these coarse priors are refined at the intermediate-resolution level by merging them with parking-stand information derived from SAR images of intermediate spatial resolution, thereby improving spatial accuracy. To ensure precise alignment and transfer of parking-stand priors, an HMPR module utilizes image registration techniques to transfer priors accurately from coarse-to-intermediate-resolution images, and further from intermediate-to-fine-resolution images. Finally, at the fine-resolution level, refined priors guide the extraction of targeted image slices from high-resolution SAR data. These slices undergo keypoint-based feature extraction and classification through a frequency-adaptive filtering network integrated with domain adaptation, enabling effective aircraft detection.

B. Coarse-Resolution Multitemporal Prior Extraction

In this phase, we present the methodology for extracting spatial priors of aircraft parking stands from coarse-resolution SAR image sequences. The goal is to exploit the high temporal frequency and broad coverage of coarse-resolution SAR to identify locations that exhibit significant backscatter variation over time due to intermittent aircraft presence. By analyzing a multitemporal stack of SAR intensity images, we derive candidate stand regions as initial priors for subsequent fine-scale analysis, as illustrated in the top of Fig. 3. Using coarse-resolution imagery is advantageous because of its frequent revisit and consistent global acquisition, which enables capturing temporal dynamics and ensures the approach is generalizable to different airports. The following describes the proposed pipeline, including motivation and technical details of each step, from multitemporal feature computation to spatial prior generation.

Multitemporal Image Stack Preparation: Let $\mathcal{J} = \{I_t(\mathbf{x})\}_{t=1}^{T}$ be a radiometrically calibrated and spatially co-registered stack

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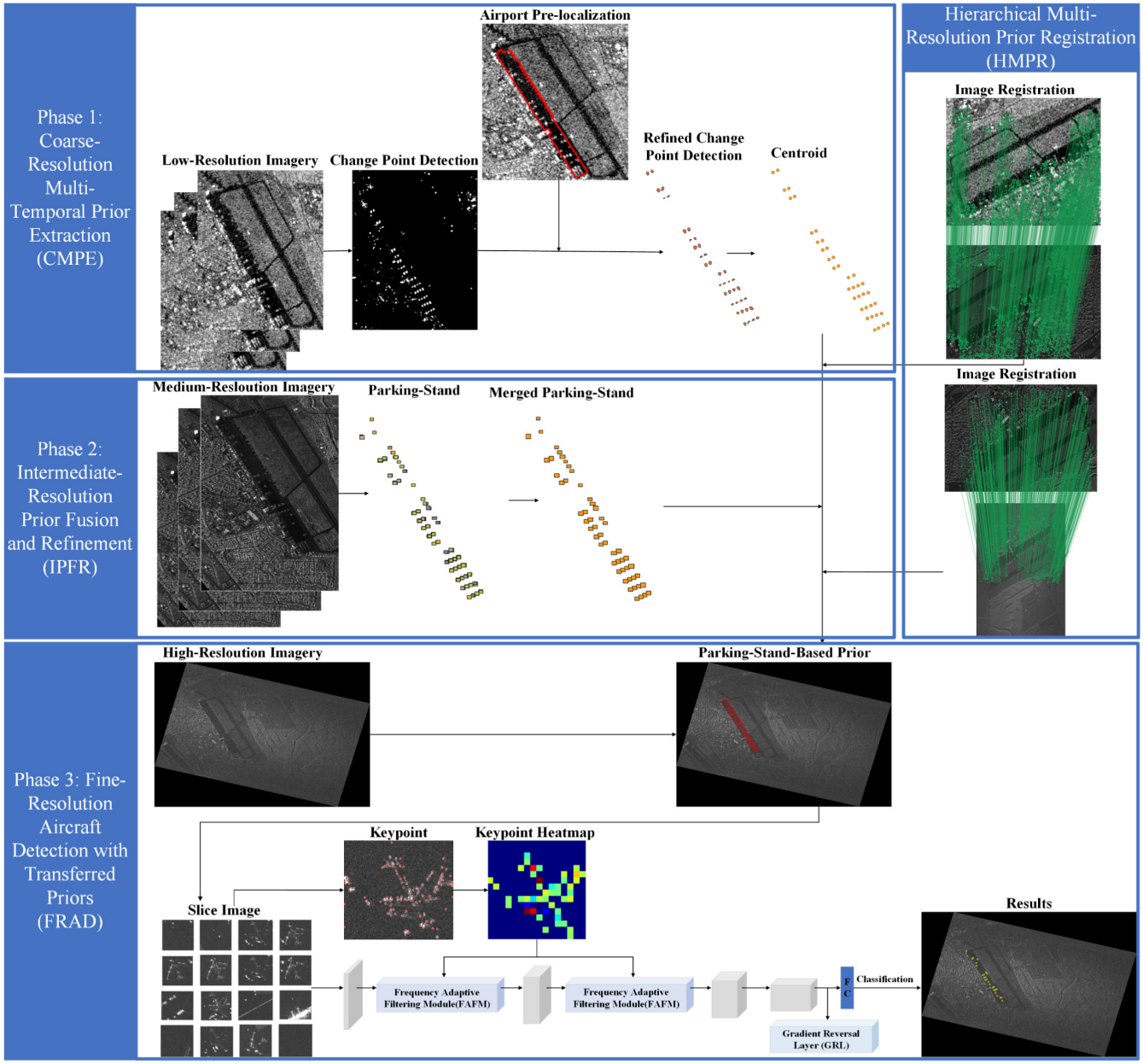

Fig. 3. Overall architecture of the HiPTA. The framework consists of three main phases: Phase 1-CMPE: Coarse-resolution, multitemporal SAR imagery is analyzed to detect temporal change points, followed by airport prelocalization and centroid extraction to generate initial parking-stand priors. Phase 2-IPFR: Intermediate-resolution SAR data is used to refine and merge parking-stand information. Hierarchical HMPR ensures spatial alignment of priors across resolutions. Phase 3-FRAD: Refined priors guide slice extraction from high-resolution images for aircraft detection using keypoint heatmaps and a domain-adaptive frequency filtering network.

of T SAR backscatter intensity images, wher $\mathbf{x} \in \Omega \subset \mathbb{R}^2$ dexes spatial ground coordinates. To suppress multiplicative speckle noise while preserving genuine temporal dynamics, we employ a 3-D nonlocal means (NLMs) filter operating jointly in spatial and temporal domains. The filtered image stack is defined as

$$\hat{I}_t(\mathbf{x}) = \frac{\sum_{\mathbf{y}\in\mathcal{N}(\mathbf{x})} \sum_{s=\max(t-L,1)}^{\min(t+L,T)} w(\mathbf{x},\mathbf{y},t,s) I_s(\mathbf{y})}{\sum_{\mathbf{y}\in\mathcal{N}(\mathbf{x})} \sum_{s=\max(t-L,1)}^{\min(t+L,T)} w(\mathbf{x},\mathbf{y},t,s)} \tag{1}$$

where the weight w is defined as

$$w(\mathbf{x},\mathbf{y},t,s) = \exp\left(-\frac{\|P_t(\mathbf{x}) - P_s(\mathbf{y})\|_2^2}{h^2}\right) \tag{2}$$

where $P_t(\mathbf{x})$ represents a local patch around the spatial position $\mathbf{x}$ in the tth image, typically of size $(2K+1)\times(2K+1)$; $\mathcal{N}(\mathbf{x})$ denotes the spatial search window centered at $\mathbf{x}$, typically larger than the patch size; L defines the temporal extent of the filtering window; and h is a data-adaptive filtering parameter determined by local image statistics, commonly set proportional to local noise variance estimates.

2) *Statistical Modeling of Temporal Variability:* For each pixel $\mathbf{x}$, we fit a p-order autoregressive (AR) model defined as

$$I_t(\mathbf{x}) = \sum_{k=1}^{p} \phi_k(\mathbf{x}) I_{t-k}(\mathbf{x}) + \varepsilon_t(\mathbf{x}), \quad t = p+1,\ldots,T \tag{3}$$

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where the AR coefficients $\{\phi_k\}$ are estimated by Burg's method. The innovation variance

$$\sigma_\varepsilon^2(\mathbf{x}) = \frac{1}{T-p}\sum_{t=p+1}^{T}\varepsilon_t^2(\mathbf{x}) \tag{4}$$

quantifies the component of backscatter variability unexplained by short-term correlations, thus effectively highlighting episodic events such as transient aircraft arrivals.

To robustly characterize temporal variability, we further incorporate two complementary statistical descriptors. First, the temporal standard deviation is defined as

$$\sigma(\mathbf{x}) = \sqrt{\frac{1}{T-1}\sum\nolimits_{t=1}^{T}\left(I_t(\mathbf{x}) - \bar{I}(\mathbf{x})\right)^2} \tag{5}$$

which quantifies overall intensity fluctuations over time. In addition, the Shannon entropy of the pixel intensity distribution across B histogram bins is given by

$$H(\mathbf{x}) = -\sum_{b=1}^{B} p_b(\mathbf{x})\log p_b(\mathbf{x}) \tag{6}$$

where $p_b(\mathbf{x})$ denotes the probability of pixel intensities at $\mathbf{x}$ falling into the histogram bin b. Entropy effectively captures irregularity and high-order dispersion within the pixel's temporal intensity profile.

These descriptors, along with the innovation variance, are normalized to comparable scales and subsequently fused into a composite variability measure

$$V(\mathbf{x}) = \tilde{\sigma}(\mathbf{x}) + \tilde{\sigma}_\varepsilon(\mathbf{x}) + \tilde{H}(\mathbf{x}) \tag{7}$$

where $\tilde{\sigma}(\mathbf{x})$, $\tilde{\sigma}_\varepsilon(\mathbf{x})$, and $\tilde{H}(\mathbf{x})$ denote the normalized counterparts of the descriptors defined in (4)–(6).

3) Candidate Generation via Adaptive Thresholding: To identify regions of significant temporal variability, we apply an adaptive threshold to the composite variability map $V(\mathbf{x})$. Since V exhibits a heavy-tailed distribution-caused by the sharp contrast between a stable background and dynamic parking stands-we use a percentile-based thresholding scheme rather than a fixed-value or mean-based approach. In particular, we select a threshold τ corresponding to the $(1-\rho)$th percentile of all $V(\mathbf{x})$ values, such that only the top $\rho\%$ of the most dynamic pixels are retained. In our implementation, ρ is empirically set to 5%, which provides a good balance between recall and false alarms. This yields the initial binary candidate mask

$$M_0(\mathbf{x}) = [V(\mathbf{x}) \geq \tau]. \tag{8}$$

4) Spatial Refinement and Prior Generation: A binary morphological closing operation is initially applied to the thresholded mask M_0 using a disk-shaped structuring element B_r, where the radius r corresponds to the coarse-resolution projection of a typical aircraft stand. In particular, the closed mask is obtained by

$$M_1 = (M_0 \oplus B_r) \ominus B_r \tag{9}$$

which reconnects fragmented regions, suppresses intrastand voids, and maintains clear interstand separations.

Next, the closed mask is segmented using an eight-connected component labeling algorithm. For each connected region C_k, the pixel area $A_k = |C_k|$ is calculated and compared against a threshold τ_A, defined as 0.75 times the median stand footprint derived from the training airports. Regions satisfying $A_k < \tau_A$ are discarded as noise or irrelevant clutter, resulting in the refined morphological mask

$$M_2(\mathbf{x}) = [\mathbf{x} \in C_k \wedge A_k \geq \tau_A]. \tag{10}$$

To eliminate residual detections outside operational airport areas, mask M_2 is further constrained by intersection with the binary airport layout mask $\mathcal{A}(\mathbf{x})$, which delineates ramp and apron areas, yielding

$$M_3(\mathbf{x}) = M_2(\mathbf{x}) \wedge \mathcal{A}(\mathbf{x}). \tag{11}$$

Subsequently, component analysis was performed on M_3 identifies N coherent candidate regions. For each region S_i, the polygonal boundary and centroid $\boldsymbol{\mu}_i$ are extracted to construct the final spatial prior set

$$\mathcal{P} = \left\{(S_i, \boldsymbol{\mu}_i)\right\}_{i=1}^{N}. \tag{12}$$

C. Intermediate-Resolution Prior Fusion and Refinement

In this phase, the coarse-resolution priors are fused with intermediate-resolution satellite detections to refine and enrich the set of parking stand candidates, as illustrated in the middle of Fig. 3. In contrast to coarse-resolution data, obtaining annotated bounding boxes for intermediate-resolution imagery can be more feasible due to its improved spatial detail. This procedure improves localization accuracy by leveraging higher resolution observations while preserving hypothesized stands from the coarse prior, ultimately yielding a more complete and precise set of stand candidates.

1) Overlapping Detection Consolidation: First, intermediate-resolution detection outputs (typically given as bounding boxes or polygonal regions) are examined for overlaps. Multiple detections covering the same physical stand are merged to eliminate redundancy. We employ the intersection-over-union (IoU) criterion to quantify overlap. For any two candidate bounding boxes b_i and b_j, their IoU is defined as the ratio of the area of overlap to the area of their union

$$\mathrm{IoU}(b_i, b_j) = \frac{|b_i \cap b_j|}{|b_i \cup b_j|}. \tag{13}$$

If $\mathrm{IoU}(b_i, b_j)$ exceeds a chosen threshold τ_{IoU}, default of 0.5 in this article, the two boxes are deemed to represent the same stand and are fused into one. In particular, we form a merged bounding region $b_{i\cup j}$ that minimally encloses both b_i and b_j. This merged box $b_{i\cup j}$ replaces the original overlapping boxes. By applying this operation iteratively, we obtain a consolidated set of detection regions $\{R_1, R_2, \ldots, R_M\}$, each ideally corresponding to a distinct stand candidate.

2) Spatial Containment-Based Prior Association: Next, each coarse prior location is linked to the nearest detection region using spatial containment. Let $P = \{\mathbf{p}_1, \mathbf{p}_2, \ldots, \mathbf{p}_N\}$ be the set of coarse-resolution priors, where $\mathbf{p}_i = (x_i^c, y_i^c)$ is the point location of the ith hypothesized stand.

For each prior $\mathbf{p}_i$ and each detection polygon R_j, we perform a point-in-polygon test to check if $\mathbf{p}_i$ lies inside R_j. If $\mathbf{p}_i \in R_j$, then the coarse prior and the detection are considered a

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match. In this case, $\mathbf{p}_i$ is fused with the region R_j, meaning that stand R_j is confirmed that the coarse prior and the intermediate annotation converge on the same geographic location.

3) Proximity-Based Interpolation for Unmatched Priors: Some coarse priors might lie outside all annotated boxes, due either to labeling omissions in the intermediate data or small spatial offsets. To ensure completeness, we adopt interpolation to handle such unmatched priors. Let $d(\mathbf{p}_i, R_j)$ be the minimum Geodesic distance from $\mathbf{p}_i$ to the perimeter of the box R_j. If

$$\min_{1\le j\le M} d\left(\mathbf{p}_i, R_j\right) \le \tau_d \tag{14}$$

the prior $\mathbf{p}_i$ is treated as belonging to whichever region R_{j^*} yields the minimum distance. The threshold τ_d is chosen based on expected geolocation accuracy.

If a prior $\mathbf{p}_i$ has no box within τ_d, it implies that the intermediate annotations missed a region that the coarse prior deemed potentially active. We create a new bounding box around $\mathbf{p}_i$ by employing a shape interpolation approach. We then compute the average bounding box width and height from the nearest k regions, then center a new box at $\mathbf{p}_i$ using these averaged dimensions. Formally, if $\mathcal{N}_k(\mathbf{p}_i)$ is the set of the nearest k boxes to $\mathbf{p}_i$, each box R_j has bounding extents $[x^j_{\min}, y^j_{\min}, x^j_{\max}, y^j_{\max}]$ from which we can derive lengths $l_j = x^j_{\max} - x^j_{\min}$ and widths $w_j = y^j_{\max} - y^j_{\min}$. The average length and width are

$$\bar{\ell}\frac{1}{|\mathcal{N}_k(\mathbf{p}_i)|}\sum_{R_j\in\mathcal{N}_k(\mathbf{p}_i)} \ell_j, \bar{w} = \frac{1}{|\mathcal{N}_k(\mathbf{p}_i)|}\sum_{R_j\in\mathcal{N}_k(\mathbf{p}_i)} w_j. \tag{15}$$

We then define a rectangle centered on $\mathbf{p}_i$ with length $\bar{\ell}$ and width $\bar{w}$, yielding a new candidate stand box. This process preserves the coarse prior while leveraging local context from existing intermediate-level annotations.

D. FRAD With Transferred Priors

In this phase, the results obtained from lower resolution analyses are transferred to the fine-resolution SAR domain and used to guide target localization, as illustrated at the bottom of Fig. 3. Each prior provides an approximate location where an aircraft is expected to be stationary. Using these transferred priors, we crop corresponding candidate image patches from the fine-resolution SAR imagery, centered on the prior coordinates. This yields a set of high-detail test patches that are highly likely to contain the target of interest. By restricting detection to these candidate regions, the search space is greatly reduced, allowing the model to focus on fine-scale features of the aircraft without distraction from irrelevant areas. Each fine-resolution patch is sized to comfortably encompass the potential aircraft and some surrounding context, ensuring that the object lies well within the patch boundaries.

To train a robust classifier for the fine-resolution patches, we leverage mid-resolution SAR imagery and the fused parking-stand priors from earlier stages. In particular, for each fused prior location, we extract a patch from the mid-resolution SAR image, using the same geographical area as the fine-resolution test patch. These mid-resolution patches serve as training samples and are assigned binary labels based on aircraft presence. Patches confirmed to contain an aircraft

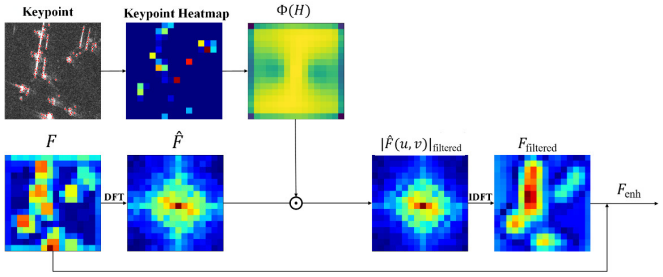

Fig. 4. Structure of the proposed FAFM. A keypoint heatmap and frequency-coordinate maps are processed by a lightweight network to generate an adaptive mask, which is applied to the magnitude spectrum of the Fourier-transformed feature map; after inverse transform, the filtered features are fused with the original features to produce the final enhanced output.

are labeled as foreground, while those with no aircraft are labeled as background. Note that although both training and test patches originate from SAR sensors, the difference in resolution and sensor characteristics introduces a domain shift between the mid-resolution training data and fine-resolution test data. We address this shift through a domain adaptation strategy described later. To classify each candidate patch as aircraft or background, we propose a convolutional neural network termed the FAFNet. The network is designed to exploit the transferred prior information for improved feature extraction and to produce domain-invariant features for generalization to fine-resolution data.

FAFNet integrates spatial and frequency domain processing with domain adaptation. It first extracts initial spatial features through a convolutional layer followed by the frequency adaptive filtering module (FAFM), which transforms features into the frequency domain, applies adaptive masks informed by keypoint heatmaps from Harris–Laplace detection, and reconstructs enhanced spatial features via inverse Fourier transform, as illustrated in Fig. 4. These features undergo further convolutional refinement and adaptive pooling, producing a compact 1024-D representation for classification. Domain adaptation is achieved using a gradient reversal layer (GRL) [30] coupled with a domain classifier, training features to be discriminative for the aircraft classification task but invariant across domains, significantly improving robustness and cross-domain performance. The architecture specifically incorporates three key modules to exploit characteristic scattering signatures in SAR imagery and adapt to multiresolution scenarios.

1) Keypoint Extraction and Heatmap Generation: Given an input SAR image $I \in \mathbb{R}^{H\times W\times 3}$, we first extract salient features using a Harris–Laplace corner detector. Formally, the detector identifies a set of keypoints $\{(x_i, y_i)\}_{i=1}^N$ by computing a corner response function over the grayscale representation of I. These responses are then aggregated into a binary map

$$H'(x,y) = \sum_{i=1}^{N} \delta\left((x,y) - (x_i, y_i)\right) \tag{16}$$

where $\delta(\cdot)$ is the Dirac delta function. To obtain a continuous representation, the binary map H' is smoothed using a Gaussian kernel, and the result is normalized to produce a keypoint

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heatmap $H \in \mathbb{R}^{16\times16}$. This heatmap serves as a spatial prior that accentuates regions with high structural significance.

2) Frequency Adaptive Filtering Module: Let $F \in \mathbb{R}^{C\times L'\times W'}$ be the feature map obtained from the initial convolutional layers applied to I. The FAF module first computes the discrete Fourier transform of F

$$\hat{F}(u,v) = \sum_{x=0}^{L'-1}\sum_{y=0}^{W'-1} F(x,y)\, e^{-j2\pi\left(\frac{ux}{L'}+\frac{vy}{W'}\right)}. \tag{17}$$

Following the DFT, a polar frequency coordinate grid is explicitly generated. The polar coordinates (r,θ) are computed for each frequency component (u,v), where the radial component r and angular component θ are given by

$$r(u,v) = \frac{\sqrt{(u-u_c)^2+(v-v_c)^2}}{r_{\max}} \tag{18}$$

$$\theta(u,v) = \arctan\left(\frac{v-v_c}{u-u_c}\right) \tag{19}$$

with (u_c, v_c) denoting the coordinates of the frequency-domain center and $r_{\max}$ denoting the maximum radial distance.

Next, a learnable filter $\Phi(H)$ is generated by concatenating the polar frequency coordinate grid (r,θ) with the keypoint heatmap H, and feeding this combined representation into a small convolutional subnetwork. Formally, this process can be represented as

$$\Phi(H) = \sigma(\mathcal{C}([H,r,\theta])) \tag{20}$$

where σ denotes a sigmoid activation function, C indicates convolutional operations, and $[H,r,\theta]$ represents channelwise concatenation. This learned filter is designed to modulate the amplitude spectrum $|\hat{F}(u,v)|$

$$\left|\hat{F}(u,v)\right|_{\text{filtered}} = \left|\hat{F}(u,v)\right| \odot \Phi(H) \tag{21}$$

where $\odot$ denotes elementwise multiplication. The phase spectrum $\angle\hat{F}(u,v)$ is preserved, and an inverse DFT is applied to reconstruct the spatial feature map

$$F_{\text{filtered}}(x,y) = \mathcal{F}^{-1}\left\{\left|\hat{F}(u,v)\right|_{\text{filtered}} e^{j\angle\hat{F}(u,v)}\right\}. \tag{22}$$

The filtered feature map F_{filtered} is subsequently fused with the original feature map F through concatenation along the channel dimension, followed by convolutional processing, resulting in an enhanced spatial representation F_{enh}. This feature enhancement step ensures that the benefits of frequency adaptive filtering is incorporated while maintaining consistency with the original feature content.

3) Domain Adaptation and Loss Functions: Since FAFNet is trained on mid-resolution patches but applied to fine-resolution patches, we integrate an adversarial domain adaptation component to bridge the gap between domains. In particular, we attach a domain classifier D to the network, which operates on the feature vector f and tries to predict whether f comes from a source (mid-resolution) or target (fine-resolution) patch. The main feature extractor is trained so that the domain classifier cannot reliably distinguish between source and target domains, leading to domain-invariant features f. This concept follows the domain-adversarial training strategy, in which the feature representation is optimized to be discriminative for the task but indiscriminate with respect to the domain shift. We implement this via a GRL inserted between the feature extractor and the domain classifier. During forward propagation, the GRL performs as an identity function; however, during back-propagation, it reverses the gradient by multiplying it by -1, effectively causing the feature extractor to maximize the domain classification loss while minimizing the primary classification loss. Let $y \in \{0,1\}$ be the class label for a given patch ($y = 1$ for aircraft-present and $y = 0$ for background) and let $d \in \{0,1\}$ be the domain label ($d = 0$ for source/mid-res and $d = 1$ for target/fine-res). FAFNet has two output branches: a classification head $C(f)$ and a domain head $D(f)$.

The classification head $C(f)$ is a fully connected network that outputs a logit, which is passed through a sigmoid $\sigma(\cdot)$ to produce the estimated probability that the patch contains an aircraft

$$\hat{y} = \sigma(C(f)). \tag{23}$$

The domain head $D(f)$ similarly produces domain probability

$$\hat{d} = \sigma(D(f)). \tag{24}$$

The network is trained using a joint loss function defined as

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{dom}} \tag{25}$$

where $\mathcal{L}_{\text{cls}}$ is the aircraft classification loss and $\mathcal{L}_{\text{dom}}$ is the domain classification loss. Classification loss $\mathcal{L}_{\text{cls}}$ is the binary cross-entropy between y and $\hat{y}$ over the source domain training samples

$$\mathcal{L}_{\text{cls}} = -\frac{1}{N_s}\sum_{i=1}^{N_s}\left[y_i \log\hat{y}_i + (1-y_i)\log(1-\hat{y}_i)\right] \tag{26}$$

where N_s is the number of labeled source (mid-resolution) patches in the batch. Domain loss $\mathcal{L}_{\text{dom}}$ is defined similarly as a binary cross-entropy on domain labels for all samples in the batch

$$\mathcal{L}_{\text{dom}} = -\frac{1}{N_s+N_t}\sum_{j=1}^{N_s+N_t}\left[d_j \log\hat{d}_j + (1-d_j)\log\left(1-\hat{d}_j\right)\right] \tag{27}$$

where N_t is the number of unlabeled target patches in the batch. During backpropagation, the gradient from $\mathcal{L}_{\text{dom}}$ is reversed by the GRL before updating the feature extraction layers, meaning the feature extractor is effectively minimizing $-\mathcal{L}_{\text{dom}}$ while the domain classifier is minimizing $\mathcal{L}_{\text{dom}}$. This adversarial training process drives the feature distributions f for mid- and fine-resolution patches to converge.

E. Hierarchical Multiresolution Prior Registration

Aircraft parking-stand priors derived from lower resolution, multitemporal SAR imagery are initially expressed in geospatial form, commonly stored as coordinates in the geographic coordinate system. However, transferring these priors across resolutions and imaging platforms introduces nonnegligible spatial misalignments due to differences in sensor geometry, spatial resolution, and map projections. To enable precise alignment and use of prior information in fine-resolution SAR imagery, we propose HMPR involving pixel-space projection

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TABLE I
DETAILED DATASET PARAMETERS IN OUR EXPERIMENT

Satellite	Band	Mode	Product Level	Polarization	Spatial Resolution	Number of Images per Airport	Temporal Coverage
Sentinel-1	C-band	IW mode	Level-1	VV	20 m	60	2016-2020
Gaofen-3	C-band	Spotlight mode	Level-1	VV+HH	1 m	10	2020-2021
Qilu-1	Ku-band	Spotlight mode	Level-1	HH	0.5 m	1	2022

and geometric warping based on image-domain homographies, as illustrated in the right of Fig. 3. Let each prior in the coarse-resolution SAR image domain I^c be described by its geographic coordinate (λ_i, ϕ_i), where λ_i and ϕ_i denote the longitude and latitude, respectively. The transformation process consists of three primary steps.

1) Geospatial-to-Pixel Conversion in Lower Resolution Domain: Each geospatial prior point is first mapped to its corresponding image-space coordinate (u_i^c, v_i^c) using the affine geotransform $\mathbf{T}_c$ associated with the coarse-resolution image

$$\begin{bmatrix} u_i^c \\ v_i^c \\ 1 \end{bmatrix} = \mathbf{T}_c^{-1} \begin{bmatrix} \lambda_i \\ \phi_i \\ 1 \end{bmatrix} \tag{28}$$

where $\mathbf{T}_c \in \mathbb{R}^{3\times3}$ is constructed from the image's metadata. This step produces a pixel-domain representation of prior locations.

2) Cross-Resolution Image Warping via Learned Homography: To transfer prior points from the coarse domain $\mathcal{I}^c$ to the fine-resolution image domain $\mathcal{I}^f$, we compute a homography matrix $\mathbf{H} \in \mathbb{R}^{3\times3}$ such that

$$\begin{bmatrix} u_i^f \\ v_i^f \\ 1 \end{bmatrix} = \mathbf{H} \begin{bmatrix} u_i^c \\ v_i^c \\ 1 \end{bmatrix} \tag{29}$$

where (u_i^f, v_i^f) are the corresponding pixel coordinates in $\mathcal{I}^f$. The homography $\mathbf{H}$ is estimated using the pretrained MatchAnything model [31], a universal cross-modality matcher that extracts high-confidence correspondences between paired SAR or cross-modal image regions. Unlike traditional georeferencing approaches, no ground control points (GCPs) or explicit tie-point labels are required. Instead, reliable patch correspondences are identified in a self-supervised manner across spatial scales, and $\mathbf{H}$ is solved via least-squares minimization of the reprojection error.

3) Pixel-to-Geospatial Recovery in Higher-Resolution Domain: After transformation, the pixel coordinate (u_i^f, v_i^f) must be reverted to geospatial form for output in a universal map-projected format. This is achieved using the affine geotransform $\mathbf{T}_f$ of the fine-resolution image

$$\begin{bmatrix} \lambda_i' \\ \phi_i' \\ 1 \end{bmatrix} = \mathbf{T}_f \begin{bmatrix} u_i^f \\ v_i^f \\ 1 \end{bmatrix}. \tag{30}$$

The resulting geographic coordinates (λ_i', ϕ_i') align accurately with the spatial reference of the finer-resolution imagery, enabling seamless overlay with auxiliary geographic layers and compatibility with various geospatial analysis tools.

III. EXPERIMENTS

A. Datasets

This study uses four independent, airport-specific multisource SAR datasets (Airports A–D). Each airport forms a separate dataset composed of Sentinel-1, Gaofen-3 (GF-3), and Qilu-1 acquisitions with complementary spatial and temporal characteristics. All training and testing are performed per-airport with nonoverlapping splits, and no cross-airport mixing or leakage is allowed. We report per-airport results across A–D to summarize overall trends. Table I summarizes the key parameters of each data source.

1) Sentinel-1: For each airport, we collected numerous Sentinel-1 C-band SAR images acquired in the interferometric wide-swath (IW) mode as Level-1 ground range detected (GRD) products. We use the vertical-vertical (VV) polarization channel from these acquisitions. Each Sentinel-1 image covers a wide swath (250 km) easily encompassing the airport area, with an approximate ground resolution of 5 × 20 m (range × azimuth). This relatively low spatial resolution is offset by the high temporal resolution: Sentinel-1 has a revisit interval of 6–12 days, enabling a long time series of dozens of images per site. In our dataset, each airport is associated with a dense multitemporal stack of Sentinel-1 images. This time-series advantage of Sentinel-1 is crucial for capturing temporal patterns and changes at the airports that a single image cannot reveal.

2) Gaofen-3: The GF-3 data used in this study are originally Level-1A single-look complex (SLC) products, which were further processed to generate radiometrically calibrated and geocoded amplitude images. Both VV and HH polarization channels were collected. These geocoded GF-3 spotlight images offer medium-to-high spatial resolution (1–5 m ground

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resolution) with a detailed depiction of airport infrastructure. Compared to Sentinel-1, GF-3 images cover a smaller area but provide sharper detail, enabling finer object localization. However, temporal coverage is relatively sparse, with only a few irregularly spaced acquisitions available per airport. As such, GF-3 serves as a high-resolution complement to the dense Sentinel-1 time series, providing precise spatial snapshots that enhance the understanding of object-level features.

3) Qilu-1: In addition, we incorporate imagery from the Qilu-1 satellite, which provides the highest spatial resolution among all data sources in our dataset. Launched in 2021, Qilu-1 is a Ku-band SAR satellite capable of acquiring submeter resolution imagery. For each airport, we obtained one Qilu-1 image acquired in high-resolution spotlight mode. These images are acquired in HH single-polarization mode. The Qilu-1 data offer ultrahigh spatial resolution (~0.5–1 m), capable of revealing fine structures and small objects on the airfield that are not discernible in C-band imagery. The tradeoff is that only one or two observations are available for each site due to the satellite's recent launch and limited acquisition opportunities, so temporal coverage is minimal, essentially a single-date snapshot for each airport. Despite the lack of time series, the Qilu-1 imagery was included to capture detailed spatial information and validate features identified in the lower resolution SAR data. Its very high spatial resolution serves as a valuable reference for object-level detail at the airport sites.

B. Implementation Details

The experiments were conducted on a Linux server equipped with a single 24-GB NVIDIA RTX 3090 GPU and using CUDA 11.7. In Phase 3, slice image patches were resized to 64 × 64 pixels, converted to grayscale, and min–max normalized to [0, 1]. Harris–Laplace keypoints were extracted with a maximum of 100 corners per image, a quality level threshold of 0.01, and a minimum interkeypoint distance of ten pixels. The extracted keypoints were represented as heatmaps, smoothed using a Gaussian kernel of size 9 × 9, normalized, and subsequently downsampled to 32 × 32 resolution. The network was trained from scratch using the Adam optimizer with an initial learning rate of 0.001 for 30 epochs. A batch size of 32 was used for both source and target domains.

C. Evaluation Metric

Detection performance is assessed using standard object detection metrics: precision, recall, $F1$-score, and average precision (AP). To ensure geometric consistency during overlap computation, all predicted and ground-truth object polygons are transformed into a unified projected coordinate reference system (CRS), specifically the corresponding universal transverse mercator (UTM) zone. This normalization ensures accurate calculation of the IoU. A detection is considered a true positive (TP) if its loU with a ground-truth polygon of the same class exceeds a predefined threshold τ. Conversely, detections with loU below τ or belonging to incorrect classes are counted as false positives (FPs). Ground-truth polygons without matching detections constitute false negatives (FNs). By varying the loU threshold τ, evaluation strictness can be adjusted; results in this study are reported at a commonly adopted threshold of $\tau = 0.5$, denoted as $AP_{0.5}$. Precision (P) and recall (R) are computed as follows:

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{31}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}. \tag{32}$$

The $F1$-score, representing the harmonic mean of precision and recall, is then given by

$$F1 = \frac{2 \times P \times R}{P + R}. \tag{33}$$

The AP provides a comprehensive measure of detection accuracy by integrating the precision–recall (PR) curve. Formally, AP is defined as the area under the PR curve

$$\mathrm{AP} = \int_0^1 P(R)\, dR \tag{34}$$

where $P(R)$ denotes precision as a function of recall R.

TABLE II
INFLUENCE OF KEY MODULES PROPOSED IN HIPTA

CMPE	IPFR	FRAD	P	R	F1	$AP_{0.5}$
√	×	×	0.8800	0.8462	0.8627	0.8354
×	√	×	0.8065	0.9615	0.8772	0.8218
√	√	×	0.8125	1.0000	0.8966	0.8503
√	×	√	1.0000	0.8462	0.9167	0.9231
×	√	√	0.9259	0.9615	0.9434	0.9224
√	√	√	0.9286	1.0000	0.9630	0.9451

TABLE III
PERFORMANCE OF EACH COMPONENT IN FAFNET

FAFM	GRL	P	R	F1	$AP_{0.5}$
×	×	0.9130	0.8077	0.8571	0.8353
×	√	0.9231	0.9231	0.9231	0.9053
√	×	0.8966	1.0000	0.9455	0.9204
√	√	0.9286	1.0000	0.9630	0.9451

D. Ablation Studies

To evaluate the effectiveness of each component within the proposed HiPTA, systematic ablation studies were conducted using data from Airport A. Each key module was isolated and evaluated individually, with results presented visually in Figs. 5–7 and quantitatively summarized in Tables II and III. In particular, we assessed the contribution of each module: CMPE, IPFR, FRAD, and HMPR.

1) Effect of CMPE: The CMPE module leverages abundant multitemporal coarse-resolution SAR imagery to generate initial spatial priors of aircraft parking stands. In particular, CMPE identifies candidate parking locations by detecting persistent changes across temporal sequences, resulting in a broad but coarse spatial map. Results in Table II clearly indicate that incorporating CMPE significantly enhances precision. Utilizing CMPE alone (without IPFR and FRAD) achieves

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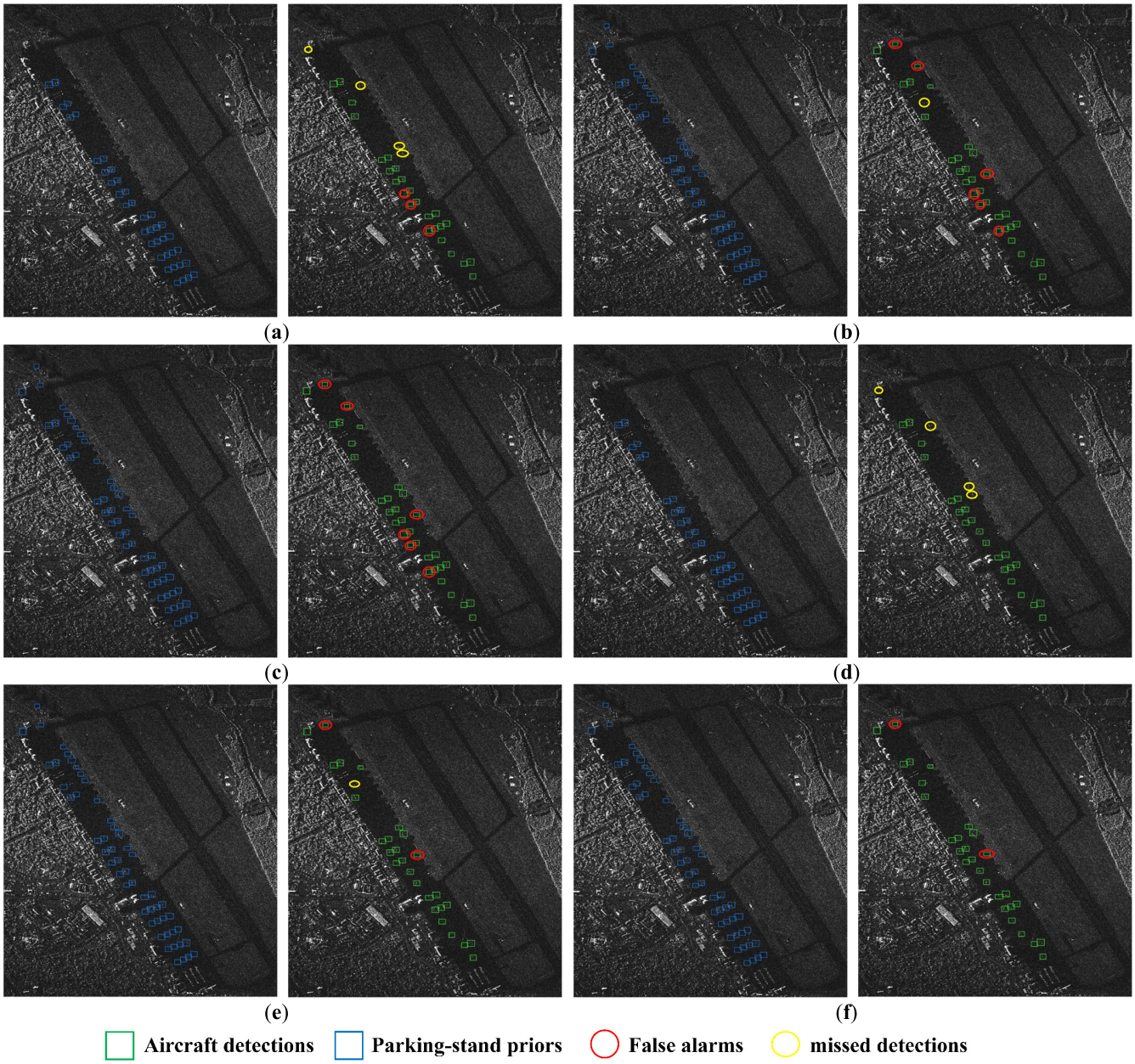

Fig. 5. Visualization results of ablation experiments. (a)–(f) Results of using the three different three-phase modules respectively, with specific configurations corresponding to rows 1–6 in Table II. In each group of pictures, the blue box in the image on the left indicates the predicted parking stands, while the green box in the image on the right represents the final detection result. The red circles denote false alarms, and the yellow circles indicate missed detections.

a precision of 88.00%, a recall of 84.62%, and an $F1$-score of 86.27%. This indicates that coarse-level priors by themselves are fairly accurate but miss some targets, as shown in Fig. 5(a). Visual inspection confirms that most aircraft lie within candidate regions, though several false alarms remain due to the absence of refined spatial priors. By contrast, without CMPE using only IPFR, the model achieves higher recall but lower precision, suggesting that intermediate-resolution data alone finds more true aircraft at the cost of more false alarms, as shown in Fig. 5(b). When CMPE is combined with IPFR, the recall reaches 100%, with precision slightly improved to 81.25%, as shown in Fig. 5(c). In other words, the coarse priors from CMPE complement the intermediate fusion, ensuring no aircraft are missed and even filtering some FPs. In summary, CMPE primarily enhanced recall, significantly increasing sensitivity to true targets with a slight precision tradeoff. Its integration effectively utilized spatial priors to minimize missed detections, thereby laying a solid foundation for subsequent modules to further eliminate false alarms.

2) Effect of IPFR: The IPFR module integrates parking stand information from intermediate-resolution SAR images with coarse priors obtained from CMPE, generating more spatially precise and comprehensive priors. This module primarily influences recall performance. As shown in Table II, employing IPFR alone (without CMPE) achieves a recall of 96.15%, significantly surpassing the 84.62% achieved by

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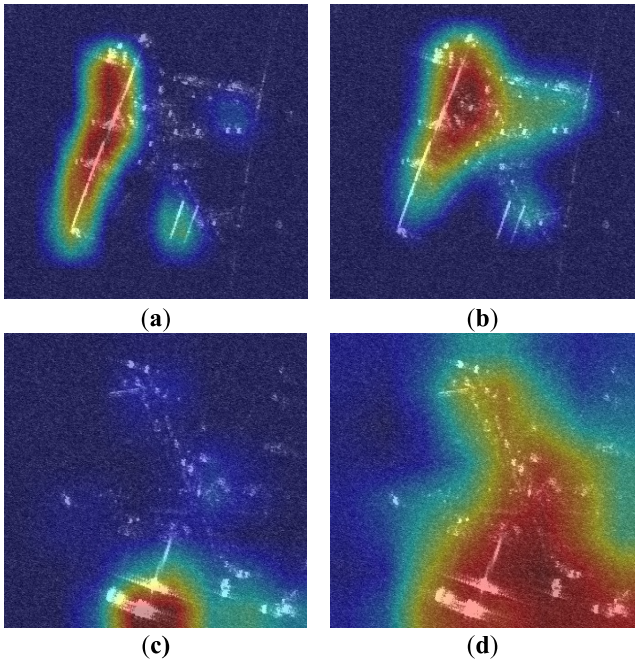

Fig. 6. Comparison visualization of the feature maps. (a) and (c) Feature maps from layers 2 to 3 of FAFNet baseline. (b) and (d) Corresponding feature maps from the FAFNet baseline with FAFM.

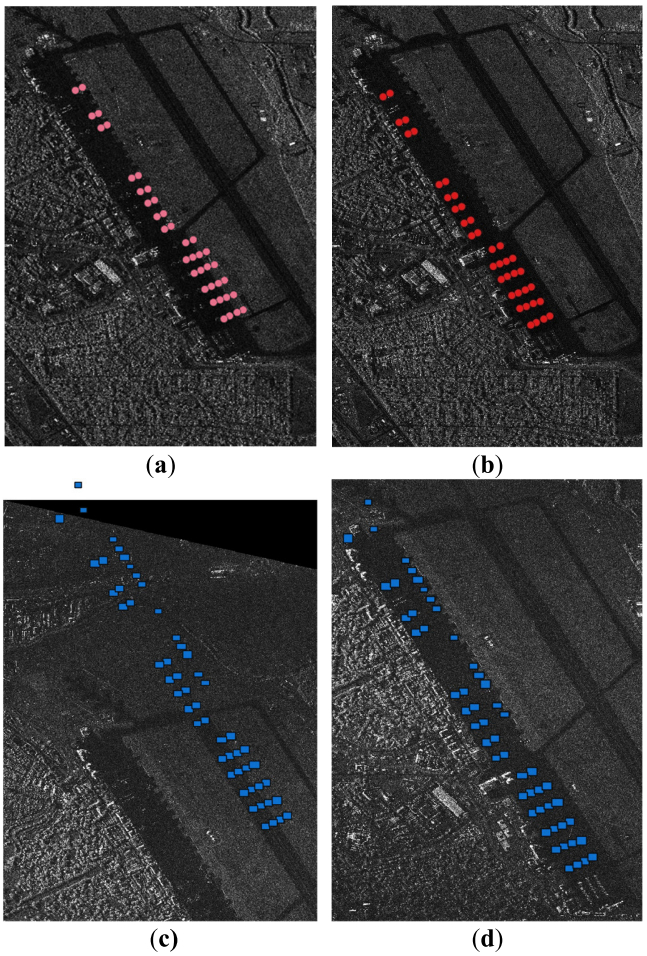

Fig. 7. Comparison visualization of the parking stands. (a) and (b) Coarse-to-intermediate-resolution transfer before and after prior registration. (c) and (d) Intermediate-to-high resolution transfer before and after prior registration.

CMPE alone. This improvement indicates that IPFR identifies additional true detections, as intermediate-resolution imagery can uncover aircraft overlooked during the coarse-resolution stage. However, using IPFR alone results in a lower precision (80.65%) compared to using only CMPE (88.00%), indicating that some intermediate-resolution priors introduced by IPFR are less reliable, leading to additional false alarms. This phenomenon is visualized in Fig. 5(b), which shows fewer missed detections compared to Fig. 5(a), albeit at the cost of increased false alarms. These additional false alarms stem from extra parking stands identified from intermediate-resolution imagery by IPFR, illustrated by the blue boxes on the left side of Fig. 5(b). When CMPE and IPFR are combined, their complementary strengths are effectively leveraged, resulting in a perfect recall of 100% and a modest precision improvement to approximately 81%, as depicted in Fig. 5(c). The perfect recall indicates that IPFR effectively addresses missed detections by exploiting intermediate-resolution data, while CMPE aids in suppressing some erroneous candidate detections. In summary, IPFR substantially reduces missed detections, effectively identifying all true aircraft. When integrated with CMPE, it yields robust priors that maximize recall while incurring only a minor tradeoff in precision.

3) Effect of FRAD: The FRAD module serves as the final detection stage, leveraging high-resolution SAR imagery to determine whether each candidate location corresponds to an aircraft. It incorporates a frequency-adaptive filtering mechanism and a GRL for domain adaptation, enabling effective exploitation of transferred priors and high-frequency spatial details for accurate target classification. The ablation results show that FRAD provides the most significant performance gains. As shown in Table II, configurations with FRAD consistently achieve significantly higher precision and $F1$ scores compared to their counterparts without FRAD. For example, applying FRAD to the IPFR-only configuration increases the $F1$ score from approximately 87.72% to 94.34%, and boosts precision from 80.65% to 92.59%, as shown in Fig. 5(e). A similar trend is observed with CMPE: while CMPE alone yields a precision of 88.00%, integrating FRAD results in a perfect 100% precision, with recall maintained at 84.62%, as shown in Fig. 5(d). The effectiveness of FRAD in suppressing FPs is evident in the full model configuration: when all three modules (CMPE + IPFR + FRAD) are included, the precision reaches 92.86% and recall remains at 100%, yielding an $F1$ score of 96.30%—a substantial improvement over the 89.66% $F1$ score achieved without FRAD (CMPE + IPFR), as illustrated in Fig. 5(f). These improvements are also visually apparent in Fig. 5, where configurations incorporating FRAD exhibit significantly fewer false alarms, confirming FRAD's effectiveness in focusing on genuine aircraft targets and suppressing spurious detections. Table III provides a detailed ablation of FRAD's internal components, highlighting the individual and joint contributions of FAFM

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and GRL. Both FAFM and GRL are shown to contribute positively to the performance of FRAD. In the absence of both components, FRAD exhibits reduced performance, with an $F1$ score of 85.71%. Introducing GRL alone, which facilitates domain

adaptation, improves the $F1$ score to 92.31%, with balanced precision and recall. This demonstrates that aligning feature distributions across source and target domains significantly enhances generalization to unseen data. Incorporating FAFM alone yields an even higher $F1$ score of 94.55%. Notably, recall reaches 100%, indicating that FAFM effectively suppresses background clutter and enhances aircraft-specific responses, thereby ensuring all targets are detected. The precision of the FAFM-only configuration is approximately 89.66%, slightly degraded due to residual FPs. However, combining FAFM with GRL compensates for this shortfall. When both components are activated, FRAD achieves optimal performance, demonstrating a strong complementary effect between FAFM and GRL. FAFM enhances the discriminability of aircraft features, ensuring no target is missed, while GRL suppresses FPs by improving domain generalization. The effectiveness of FAFM is also visually evident in Fig. 6. The feature maps from the baseline network [see Fig. 6(a) and (c)] exhibit weaker activations at aircraft locations, whereas those with FAFM [see Fig. 6(b) and (d)] display significantly enhanced responses, clearly distinguishing aircraft from background clutter. These observations highlight how FAFM guides the network to focus on genuine aircraft signatures while attenuating irrelevant background features. Overall, FRAD—particularly with the integration of FAFM and GRL—is essential for high-fidelity classification, transforming coarse and intermediate spatial priors into precise final detections by substantially reducing FPs and mitigating cross-domain discrepancies.

4) Effect of HMPR: The HMPR module ensures that priors extracted at one resolution are accurately mapped to the next resolution level. It performs hierarchical image registration from coarse to intermediate, and then to fine resolution levels, ensuring that candidate parking stand priors remain spatially aligned with the actual aircraft positions at each scale. Although HMPR does not directly contribute to precision or recall when evaluated in isolation, its influence manifests in the improved consistency and accuracy of multiresolution detections. In the absence of HMPR, transferred priors often suffer from spatial misalignment; for instance, a parking stand detected at coarse resolution may deviate by several pixels from the actual aircraft location in intermediate or fine-resolution images. Such misalignments may result in missed detections or FPs. This is clearly demonstrated in Fig. 7: prior to registration, both the coarse-to-intermediate [see Fig. 7(a)] and intermediate-to-fine [see Fig. 7(c)] transfers exhibit noticeable spatial offsets between predicted stands and actual aircraft positions. After applying HMPR, the priors are correctly aligned with the aircraft in both intermediate and fine-resolution images [see Fig. 7(b) and (d)], as evidenced by the precise overlap between the green detection boxes and blue prior locations. The experimental results show that incorporating HMPR consistently enhances detection outcomes by guiding the high-resolution classifier to focus on spatially accurate candidate regions.

TABLE IV
PER-AIRPORT PERFORMANCE OF HIPTA

Airport	P	R	F1	$AP_{0.5}$
A	0.9286	1.0000	0.9630	0.9451
B	0.9310	0.9000	0.9153	0.9117
C	0.7370	0.9680	0.8371	0.8284
D	1.0000	0.8333	0.9091	0.9167

E. Experimental Results and Analysis

1) Per-Airport Performance: To assess cross-scene robustness, HiPTA is evaluated on four independent airport-specific multisource SAR datasets covering Airports A–D. The quantitative outcomes are summarized in Table IV, and representative visualizations are provided in Fig. 8, where the columns follow the A–D ordering to aid cross-reference. Airports A and B exhibit compact aprons with two to four angled rows of parked

aircraft; Airport A further contains many recessed indented stands, whereas Airport B contains few such bays. These concentrated layouts favor the transfer of multitemporal priors to fine-resolution detection, since the priors align tightly with stand geometry and restrict the search to a small set of high-likelihood regions, which increases recall while limiting background responses. In contrast, Airports C and D span larger and more dispersed areas. Airport C combines numerous inward-recessed stands with multiple aprons and runway-adjacent zones, while Airport D displays uneven dispersion and curved stands that alter the local spatial context. In these settings, the priors still guide detections, yet clutter near indented bays and heterogeneous surroundings such as turf or adjacent buildings can trigger responses outside target footprints, and atypical placements or weak-scattering aircraft can lead to a small number of misses.

The blue boxes in Fig. 8 represent parking-stand priors estimated from the time series, and they coherently cover the ground-truth aircraft across all four airports, which indicates that the prior extraction is reliable when sufficient temporal depth is available. The green boxes produced by HiPTA on high-resolution SAR align closely with these priors, thereby concentrating computation where targets are likely to occur and suppressing broad-area background responses. Because the priors are intentionally redundant to avoid missing transient stands, a limited number of false alarms can appear in structurally complex zones; nevertheless, relative to exhaustive sliding-window scanning over full-scene panoramas, the prior-constrained strategy yields fewer spurious responses and improved efficiency.

2) Comparison With State-of-the-Art: This section compares HiPTA with representative two-stage detectors (Faster R-CNN [32], Cascade R-CNN [33]), one-stage detectors (RetinaNet [34], FSAF [35], and CenterNet [36]), and transformer-based detectors DETR [37] and DINO [38], together with their airport-detection-aided variants denoted

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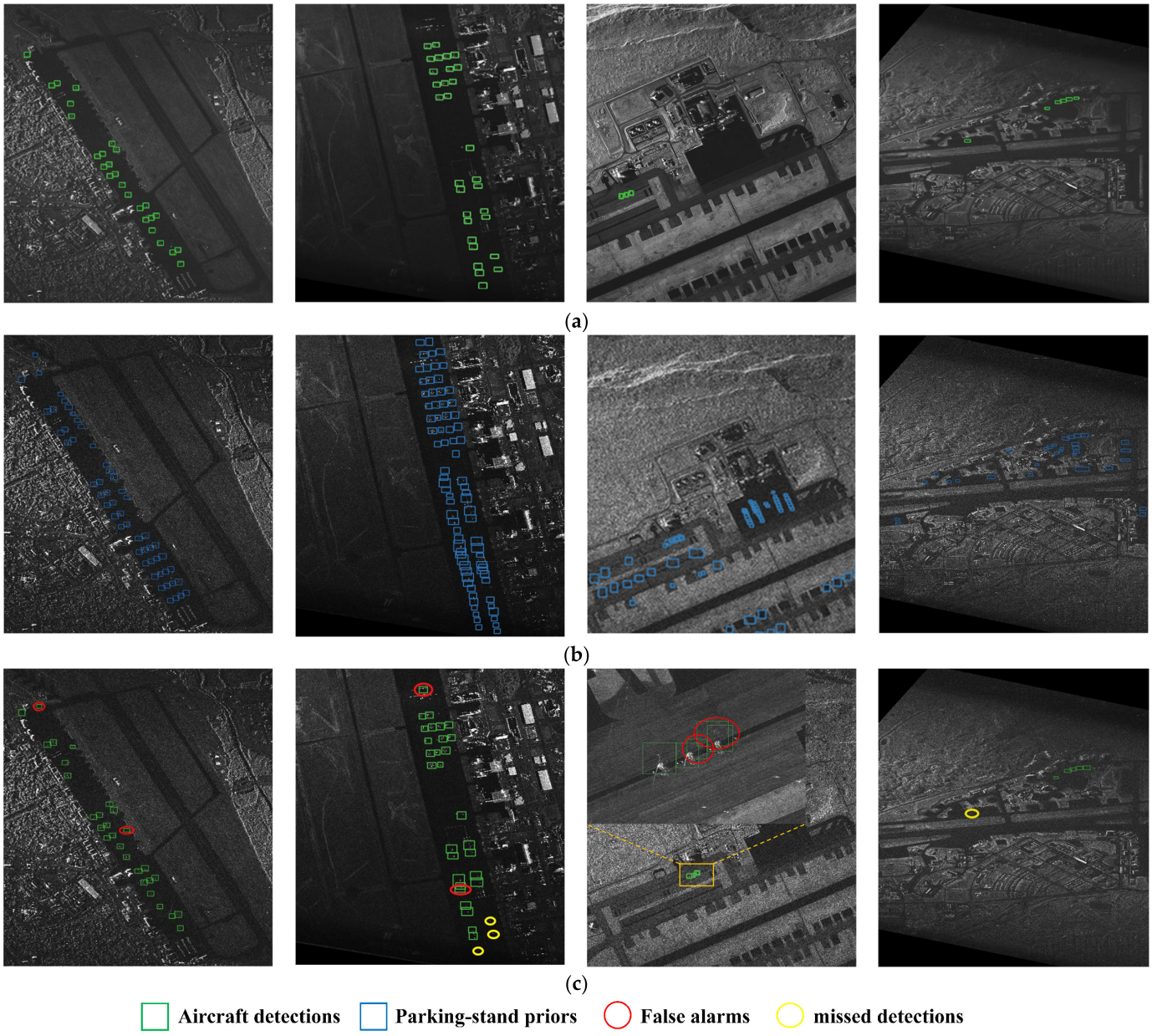

Fig. 8. Per-airport results of HiPTA. Columns: Airports A–D. Green boxes denote aircraft detections, blue boxes indicate parking-stand priors, red circles highlight false alarms, and yellow circles mark missed detections. (a) Ground-truth aircraft. (b) Parking-stand priors estimated from multitemporal SAR images. (c) Detections by HiPTA.

"w/AD." We also include an internal baseline HiPTA-Basic, which retains the hierarchical parking-stand priors but replaces the fine-resolution head with a plain head without FAFM and GRL to isolate the head's contribution. To ensure a fair evaluation, each method was equipped with a preliminary airport detection step (denoted "w/AD") that confines subsequent aircraft search to a likely airport. This AD strategy follows prior works [7], [13], [15] that leverage geospatial context to constrain detection, thereby mitigating the inefficiencies and high false-alarm rates caused by exhaustive scanning of large SAR panoramas. The detectors were trained on the Qilu-1 imagery by slicing large scenes into 1024 × 1024 windows with a 512-pixel stride, and evaluated in the same sliding-window manner.

Table V summarizes the comparative results and reveals three consistent trends. First, restricting the search to the airport region in the w/AD setting uniformly stabilizes all baselines by suppressing panorama-scale clutter, which yields a more favorable precision–recall operating point without compromising recall. Second, HiPTA-Basic, which retains hierarchical stand-level priors while using a plain head, attains higher precision than scene-level context alone, indicating that stand priors substantively reduce background-induced false alarms rather than merely reweighting proposals. Third, the full HiPTA further shifts the precision–recall curve outward relative to both CNN and recent Transformer detectors, including DETR and the stronger DINO variant, demonstrating that frequency-adaptive filtering and

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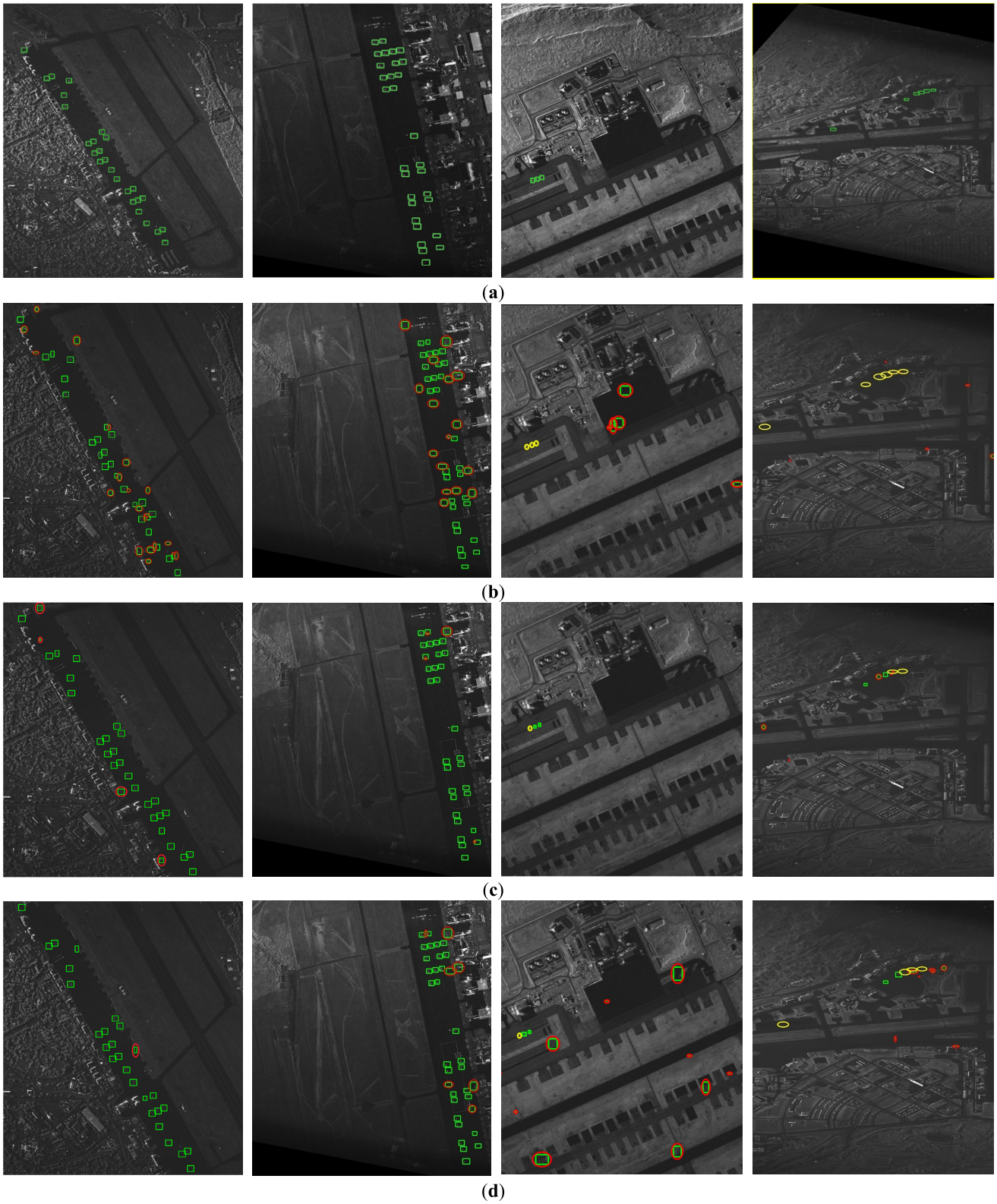

Fig. 9. Comparison of detection results from different methods across four airports. Columns: Airports A–D. Green boxes denote aircraft detections, red circles highlight false alarms, and yellow circles mark missed detections. (a) Ground truth. (b) Faster R-CNN (w/AD). (c) FSAF (w/AD). (d) CenterNet (w/AD). (e) HiPTA.

domain alignment enhance discriminability on top of the priors. The complexity columns additionally show that these gains are achieved with modest online cost per forward pass.

Fig. 9 provides a ground-truth-referenced comparison across Airports A–D. Relative to the annotations in Fig. 9(a), Faster R-CNN (w/AD) in Fig. 9(b) exhibits residual false alarms along apron edges and taxiway-adjacent clutter; FSAF (w/AD)

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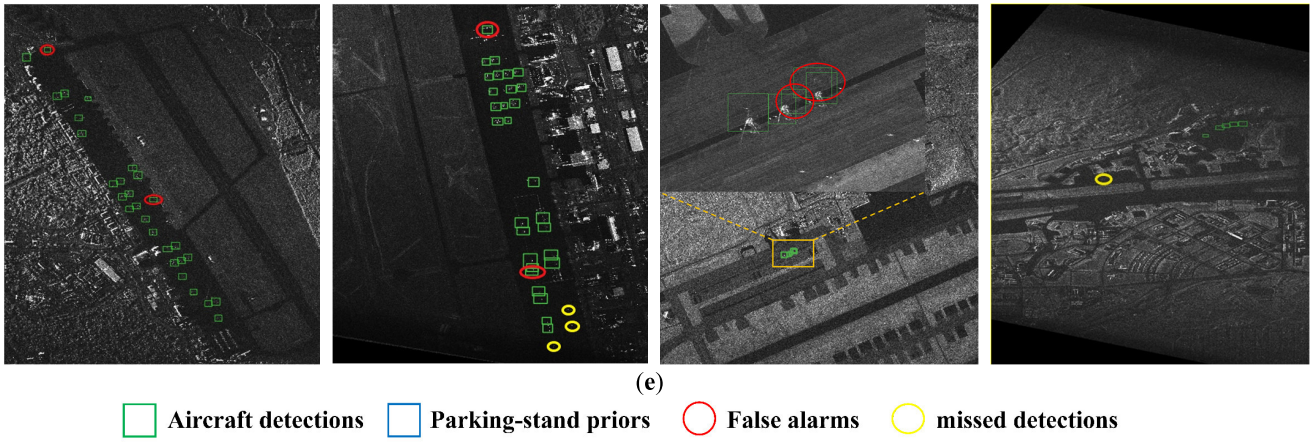

Fig. 9. *(Continued.)*Comparison of detection results from different methods across four airports. Columns: Airports A–D. Green boxes denote aircraft detections, red circles highlight false alarms, and yellow circles mark missed detections. (a) Ground truth. (b) Faster R-CNN (w/AD). (c) FSAF (w/AD). (d) CenterNet (w/AD). (e) HiPTA.

TABLE V
COMPARISON WITH STATE-OF-THE-ART METHODS AND AIRPORT-DETECTION-AIDED VARIANTS

Method	P	R	F1	$AP_{0.5}$	Params(M)	FLOPs(G)
Faster R-CNN	0.4710	0.8904	0.6161	0.7470	41.35	125.10
Cascade R-CNN	0.4393	0.7836	0.5630	0.7510	69.15	152.58
RetinaNet	0.4763	0.8871	0.6198	0.7540	36.30	117.03
FSAF	0.4883	0.9113	0.6359	0.8110	36.24	206.80
CenterNet	0.4969	0.9258	0.6467	0.8350	32.11	205.97
Faster R-CNN (w/ AD)	0.7471	0.8387	0.7903	0.8110	41.35	125.10
Cascade R-CNN (w/ AD)	0.7550	0.8358	0.7933	0.8230	69.15	152.58
RetinaNet (w/ AD)	0.8243	0.9094	0.8648	0.8240	36.30	117.03
FSAF (w/ AD)	0.8111	0.9306	0.8668	0.8930	36.24	206.80
CenterNet (w/ AD)	0.7952	0.9435	0.8630	0.8820	32.11	205.97
DETR (w/ AD)	0.7806	0.9234	0.8460	0.8573	41.55	99.80
DINO (w/ AD)	0.8084	0.9355	0.8673	0.8892	47.02	279.81
HiPTA-Basic	0.8921	0.9368	0.9139	0.8940	28.94	100.73
HiPTA	0.9104	0.9403	0.9251	0.8997	31.37	105.52

"(w/ AD)" indicates that a preliminary airport detection step is employed to restrict the search area for subsequent aircraft detection.

in Fig. 9(c) reduces some misses yet still responds to bright linear structures; and CenterNet (w/AD) in Fig. 9(d) produces dense activations in tightly spaced or heterogeneous zones, yielding additional FPs. By contrast, HiPTA in Fig. 9(e) concentrates responses within plausible stand regions, showing visibly fewer false alarms while maintaining low miss rates, including under the dispersed parking patterns at Airport D. These qualitative observations accord with the quantitative trends in Table V, indicating that prior-guided search coupled with fine-resolution refinement attains a more favorable precision–recall balance than detectors equipped solely with airport-detection aid.

IV. CONCLUSION

This work presented HiPTA, a hierarchical prior transfer framework that fuses multitemporal and multiresolution SAR information in a coarse-to-fine pipeline. The method converts long-horizon temporal evidence into spatial occupancy priors, refines them across resolutions, and guides a frequency-adaptive fine-resolution detector to focus on likely locations while suppressing background responses. The experiments on four heterogeneous airports and multiple sensors show consistent improvements in the balance between precision and recall and robust performance across scenes. The results indicate that multitemporal imagery is operationally useful when distilled into stand-level priors rather than treated as an auxiliary cue. Ablation studies confirm that each stage contributes synergistically: multitemporal priors expand recall, cross-resolution fusion sharpens localization, and the domain-adaptive fine detector markedly reduces false alarms.

Several limitations remain. The multistage design increases engineering effort and can accumulate errors, which may complicate deployment. Apron usage and gate assignment evolve over time, so fixed or infrequently updated parking-stand priors can drift from current conditions and misalign with instantaneous observations. To address these issues, future work will use self-supervised learning on long SAR time series to learn stand-level fields that capture occupancy frequency, persistence, recency, turnover rhythms, and seasonality, and will embed these summaries within the detector as differentiable, uncertainty-aware, and time-indexed prior maps. This

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will enable end-to-end optimization, dynamic weighting of recent and historical cues, online updates, and suppression of stale signals.

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